

28 Supersymmetric Versions of the Standard Model

28.5 The Sector of Zero Baryon and Lepton Number

Despite the large number of parameters of the supersymmetric standard model, in some contexts it is surprisingly predictive. This is true in particular when we consider the scalar fields whose vacuum expectation values spontaneously break the $SU(2) \cdot U(1)$ gauge symmetry. In this section we will consider these scalars, along with other fields of zero baryon and lepton number: neutral scalars odd under charge conjugation, charged scalars, and the fermionic superpartners of these scalars and of the $W^\pm$ and Z^0 .

It is a crucial requirement for supersymmetric versions of the standard model that they should contain scalar doublet ‘Higgs’ superfields with the right mass and interaction parameters to account for the breakdown of the $SU(2) \cdot U(1)$ gauge group of the electromagnetic and weak interactions. We saw in Section 28.1 that at least two left-chiral scalar doublets are needed to give mass to the quarks of both charge $2e/3$ and $-e/3$ and the charged leptons, while we found in Section 28.2 that two doublets are just what is needed to bring the $SU(3)$, $SU(2)$, and $U(1)$ gauge couplings together at some very high energy. We have therefore assumed that there are two left-chiral scalar $SU(2)$ doublets

$$H_1 = \begin{pmatrix} H_1^0 \\ H_1^- \end{pmatrix}, \quad H_2 = \begin{pmatrix} H_2^+ \\ H_2^0 \end{pmatrix}. \quad (28.5.1)$$

These have $SU(2)$ and $U(1)$ D -terms (27.4.7) given (assuming zero Fayet–Iliopoulos constant $\xi_{U(1)}$) by

$$D = \frac{g}{2} \left(\mathcal{H}_1^\dagger \boldsymbol{\tau} \mathcal{H}_1 \right) + \frac{g}{2} \left(\mathcal{H}_2^\dagger \boldsymbol{\tau} \mathcal{H}_2 \right), \quad (28.5.2)$$

$$D_y = \frac{g'}{2} \left(\mathcal{H}_1^\dagger \mathcal{H}_1 \right) - \frac{g'}{2} \left(\mathcal{H}_2^\dagger \mathcal{H}_2 \right), \quad (28.5.3)$$

where $\mathcal{H}_{1,2}$ are the scalar components of the superfield doublets $H_{1,2}$ and $\boldsymbol{\tau}_r$ are the Pauli matrices, with $\tau_r^2 = \mathbf{1}$. As shown in Eq. (27.4.9), in renormalizable theories this gives the D -term contribution to the scalar field potential

$$\begin{aligned} V_D &= \frac{1}{2} D^2 + \frac{1}{2} D_y^2 \\ &= \frac{g^2}{8} \left[\left(\mathcal{H}_1^\dagger \boldsymbol{\tau} \mathcal{H}_1 \right) + \left(\mathcal{H}_2^\dagger \boldsymbol{\tau} \mathcal{H}_2 \right) \right]^2 + \frac{g'^2}{8} \left[\left(\mathcal{H}_1^\dagger \mathcal{H}_1 \right) - \left(\mathcal{H}_2^\dagger \mathcal{H}_2 \right) \right]^2. \end{aligned} \quad (28.5.4)$$

This can be put in a more convenient form by using the relation

$$(\boldsymbol{\tau})_{i\ell} \cdot (\boldsymbol{\tau})_{kj} = 2\delta_{ij}\delta_{k\ell} - \delta_{i\ell}\delta_{kj}. \quad (28.5.5)$$

(To prove this, use rotational invariance to show that $\delta_{ij}\delta_{k\ell}$ may be expressed as a linear combination of $(\boldsymbol{\tau})_{i\ell} \cdot (\boldsymbol{\tau})_{kj}$ and $\delta_{i\ell}\delta_{kj}$, and calculate the coefficients by taking traces on the indices i, j and i, ℓ .) In this way, we may rewrite the D -term part of the scalar field potential as

$$V_D = \frac{g^2}{2} \left| \mathcal{H}_1^\dagger \mathcal{H}_2 \right|^2 + \frac{g^2 + g'^2}{8} \left[\left(\mathcal{H}_1^\dagger \mathcal{H}_1 \right) - \left(\mathcal{H}_2^\dagger \mathcal{H}_2 \right) \right]^2. \quad (28.5.6)$$

As mentioned in Section 28.1, there is just one possible renormalizable term in the superpotential for these two left-chiral doublets, of the form

$$f(H_1, H_2) = \mu (H_1^T e H_2), \quad (28.5.7)$$

where μ is a constant with the dimensions of a mass and e is the antisymmetric matrix $i\tau_2$. According to Eq. (27.4.9), this gives an additional contribution to the scalar field potential

$$\begin{aligned} V_\mu &= \sum_r \left| \frac{\partial f(\mathcal{H}_1, \mathcal{H}_2)}{\partial \mathcal{H}_{1r}} \right|^2 + \sum_r \left| \frac{\partial f(\mathcal{H}_1, \mathcal{H}_2)}{\partial \mathcal{H}_{2r}} \right|^2 \\ &= |\mu|^2 \left[(\mathcal{H}_1^\dagger \mathcal{H}_1) + (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]. \end{aligned} \quad (28.5.8)$$

For $\mu \neq 0$ the potential $V_D + V_\mu$ evidently has a minimum value of zero, reached at the unique point $\mathcal{H}_1 = \mathcal{H}_2 = 0$. With just these terms in the potential, $SU(2) \cdot U(1)$ as well as supersymmetry is not spontaneously broken. (The case $\mu = 0$ is not much better; there is a continuous infinity of vacuum states with supersymmetry unbroken and with $SU(2) \cdot U(1)$ broken down to electromagnetic gauge invariance with all possible strengths, including zero.) This is one more example of the general difficulty, already seen in Section 28.3, of formulating realistic theories in which supersymmetry is spontaneously broken within the standard model.

Under the assumption of the previous section that supersymmetry is violated in the effective Lagrangian only by superrenormalizable terms, the most general such supersymmetry-breaking term involving the scalar doublets is of the form

$$V_m = m_1^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + m_2^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) + \text{Re} \{ B\mu (\mathcal{H}_1^T e \mathcal{H}_2) \},$$

where m_1^2 and m_2^2 are real parameters (not necessarily positive) and $B\mu$ is a parameter of arbitrary phase. We will adjust the over-all phase of the superfields H_1 and H_2 so that $B\mu$ is real and positive, and so

$$V_m = m_1^2 (\mathcal{H}_1^\dagger \mathcal{H}_1) + m_2^2 (\mathcal{H}_2^\dagger \mathcal{H}_2) + B\mu \text{Re} (\mathcal{H}_1^T e \mathcal{H}_2). \quad (28.5.9)$$

The total scalar potential in the tree approximation is then

$$\begin{aligned} V &= V_D + V_\mu + V_m \\ &= \frac{g^2}{2} |\mathcal{H}_1^\dagger \mathcal{H}_2|^2 + \frac{g^2 + g'^2}{8} \left[(\mathcal{H}_1^\dagger \mathcal{H}_1) - (\mathcal{H}_2^\dagger \mathcal{H}_2) \right]^2 \\ &\quad + (m_1^2 + |\mu|^2) (\mathcal{H}_1^\dagger \mathcal{H}_1) + (m_2^2 + |\mu|^2) (\mathcal{H}_2^\dagger \mathcal{H}_2) \\ &\quad + B\mu \text{Re} (\mathcal{H}_1^T e \mathcal{H}_2). \end{aligned} \quad (28.5.10)$$

Note in particular that μ^2 , m_1^2 , and m_2^2 only appear in the combinations $m_1^2 + |\mu|^2$ and $m_2^2 + |\mu|^2$.

There is one condition on the supersymmetry-breaking parameters m_i^2 , derived from the requirement that the potential should be bounded below. For scalar fields going to infinity in generic directions, the potential is dominated by the quartic terms V_D , which

are positive. There are special directions in which V_D vanishes: those for which (up to an $SU(2) \cdot U(1)$ gauge transformation)

$$\mathcal{H}_1 = \begin{pmatrix} \phi \\ 0 \end{pmatrix}, \quad \mathcal{H}_2 = \begin{pmatrix} 0 \\ \phi \end{pmatrix},$$

with ϕ an arbitrary complex quantity. For such directions, $V = (2|\mu|^2 + m_1^2 + m_2^2)|\phi|^2 - B\mu\phi^2$, so (since $B\mu$ has been defined to be positive) in order for this not to go to $-\infty$ as $\phi \rightarrow +\infty$, it is necessary that

$$2|\mu|^2 + m_1^2 + m_2^2 \geq B\mu. \quad (28.5.11)$$

We wish to look for a minimum of the potential at which electromagnetic gauge invariance is not broken, so let us consider the behavior of the potential as a function of the neutral scalar fields, with the charged scalar fields set equal to zero. In this case Eq. (28.5.10) gives the potential of the neutral scalars as

$$V^N = \frac{g^2 + g'^2}{8} [|\mathcal{H}_1^0|^2 - |\mathcal{H}_2^0|^2]^2 + (m_1^2 + |\mu|^2)|\mathcal{H}_1^0|^2 + (m_2^2 + |\mu|^2)|\mathcal{H}_2^0|^2 - B\mu \operatorname{Re}(\mathcal{H}_1^0 \mathcal{H}_2^0). \quad (28.5.12)$$

To find a stationary point, we expand V^N around the constant values $\mathcal{H}_i^0 = v_i$, writing

$$\mathcal{H}_i^0 = v_i + \varphi_i. \quad (28.5.13)$$

To second order in the φ_i , Eq. (28.5.12) gives

$$\begin{aligned} V_{\text{quad}}^N = & \frac{g^2 + g'^2}{4} (|v_1|^2 - |v_2|^2) [2 \operatorname{Re}(v_1^* \varphi_1 - v_2^* \varphi_2) + |\varphi_1|^2 - |\varphi_2|^2] \\ & + \frac{g^2 + g'^2}{2} [\operatorname{Re}(v_1^* \varphi_1 - v_2^* \varphi_2)]^2 \\ & + (m_1^2 + |\mu|^2) (2 \operatorname{Re} v_1^* \varphi_1 + |\varphi_1|^2) \\ & + (m_2^2 + |\mu|^2) (2 \operatorname{Re} v_2^* \varphi_2 + |\varphi_2|^2) \\ & - B\mu \operatorname{Re}(v_1 \varphi_2 + v_2 \varphi_1 + \varphi_1 \varphi_2) + \text{constant}. \end{aligned} \quad (28.5.14)$$

For the v_i to be equilibrium values of the fields, the terms of first order in the φ_i must vanish

$$(m_1^2 + |\mu|^2)v_1^* + \frac{g^2 + g'^2}{4} (|v_1|^2 - |v_2|^2)v_1^* - \frac{1}{2}B\mu v_2 = 0, \quad (28.5.15)$$

$$(m_2^2 + |\mu|^2)v_2^* + \frac{g^2 + g'^2}{4} (|v_2|^2 - |v_1|^2)v_2^* - \frac{1}{2}B\mu v_1 = 0. \quad (28.5.16)$$

Without changing the over-all phase of the φ_i , we may adjust their relative phases so that v_1 is real. Then Eqs. (28.5.15) and (28.5.16) show that v_2 is also real, so that these equations become

$$(m_1^2 + |\mu|^2)v_1 + \frac{g^2 + g'^2}{4} (v_1^2 - v_2^2)v_1 - \frac{1}{2}B\mu v_2 = 0, \quad (28.5.17)$$

$$(m_2^2 + |\mu|^2)v_2 + \frac{g^2 + g'^2}{4} (v_2^2 - v_1^2)v_2 - \frac{1}{2}B\mu v_1 = 0. \quad (28.5.18)$$

These conditions may be used to express the mass parameters in the potential in terms of the convenient quantities

$$\tan \beta \equiv v_2/v_1, \quad (28.5.19)$$

$$m_Z^2 \equiv \frac{1}{2}(g^2 + g'^2)(v_1^2 + v_2^2), \quad (28.5.20)$$

and

$$m_A^2 \equiv 2|\mu|^2 + m_1^2 + m_2^2. \quad (28.5.21)$$

(The parameter m_Z is the mass of the Z vector boson.¹ We will soon see that m_A is the mass of one of the physical scalars.) Multiplying Eqs. (28.5.17) and (28.5.18), respectively, by v_2 and v_1 and taking the sum and difference gives

$$B\mu = m_A^2 \sin 2\beta, \quad (28.5.22)$$

and

$$m_1^2 - m_2^2 = -(m_A^2 + m_Z^2) \cos 2\beta, \quad (28.5.23)$$

which with Eq. (28.5.21) yields

$$\begin{aligned} m_1^2 + |\mu|^2 &= \frac{1}{2}m_A^2 - \frac{1}{2}(m_A^2 + m_Z^2) \cos 2\beta, \\ m_2^2 + |\mu|^2 &= \frac{1}{2}m_A^2 + \frac{1}{2}(m_A^2 + m_Z^2) \cos 2\beta. \end{aligned} \quad (28.5.24)$$

With linear terms cancelling, the quadratic part (28.5.14) of the neutral scalar potential may then be written

$$\begin{aligned} V_{\text{quad}}^N &= \frac{g^2 + g'^2}{4}(v_1^2 - v_2^2) [|\varphi_1|^2 - |\varphi_2|^2] + \frac{g^2 + g'^2}{2} [\text{Re}(v_1\varphi_1 - v_2\varphi_2)]^2 \\ &\quad + (m_1^2 + |\mu|^2)|\varphi_1|^2 + (m_2^2 + |\mu|^2)|\varphi_2|^2 - B\mu \text{Re}(\varphi_1\varphi_2) + \text{constant} \\ &= \frac{1}{2}m_Z^2 \cos 2\beta [|\varphi_1|^2 - |\varphi_2|^2] + m_Z^2 [\text{Re}(\cos \beta \varphi_1 - \sin \beta \varphi_2)]^2 \\ &\quad + \frac{1}{2}m_A^2 (|\varphi_1|^2 + |\varphi_2|^2) - \frac{1}{2}(m_A^2 + m_Z^2) \cos 2\beta [|\varphi_1|^2 - |\varphi_2|^2] \\ &\quad - m_A^2 \sin 2\beta \text{Re}(\varphi_1\varphi_2) + \text{constant}. \end{aligned} \quad (28.5.25)$$

We see from Eq. (28.5.25) that the real and imaginary parts of the φ_i are decoupled. (This is because the potential (28.5.12) is invariant under a charge conjugation or CP transformation $\varphi_i \rightarrow \varphi_i^*$.) The mass-squared matrix of the imaginary parts of the φ_i is

$$M_{\text{Im}\varphi}^2 = \begin{pmatrix} \frac{1}{2}m_A^2(1 - \cos 2\beta) & \frac{1}{2}m_A^2 \sin 2\beta \\ \frac{1}{2}m_A^2 \sin 2\beta & \frac{1}{2}m_A^2(1 + \cos 2\beta) \end{pmatrix}. \quad (28.5.26)$$

The determinant vanishes, so one eigenvalue is zero, and the other equals the trace, which is just m_A^2 . The zero-mass scalar is of course the neutral Goldstone boson associated with the spontaneous breakdown of $SU(2) \cdot U(1)$ to electromagnetic gauge invariance, and as

¹There is a difference of a factor of 2 between the formulas for m_Z^2 given by Eqs. (28.5.20) and (21.3.30), due to the fact that the scalar fields are normalized differently here and in Section 21.3.

discussed in Chapter 21 it is eliminated by the Higgs mechanism. As promised, m_A is the mass of one of the physical scalars, the non-Goldstone boson with C negative. This shows that for the field value $\varphi_i = v_i$ to be at least a local minimum of the potential, the parameter m_A^2 defined by Eq. (28.5.21) must be positive. The condition (28.5.11) for good behavior at large field strengths shows that Eq. (28.5.22) has a solution here for β in the range $0 \leq \beta \leq \pi/2$.

In particular, if $B\mu = 0$ and $0 < \beta < \pi/2$, then Eq. (28.5.22) shows that $m_A = 0$. In this case, the particle A is the Goldstone boson of a $U(1)$ Peccei–Quinn symmetry [4] of the potential (28.5.12) under an equal phase change of $\mathcal{H}_1^0$ and $\mathcal{H}_2^0$, which for $v_1 \neq 0$ and $v_2 \neq 0$ is spontaneously broken without leaving any combination of this and the electroweak $U(1)$ symmetry unbroken. This is the original version of the axion [25], which as we saw in Section 23.6 acquires only a small mass from the Yukawa interactions of the scalars with the quarks and is experimentally ruled out. Thus we can conclude that $B\mu$ definitely does not vanish.

The elements of the mass-squared matrix for the real scalars are given by Eq. (28.5.25) as

$$\begin{aligned} (M_{\text{Re}\varphi}^2)_{11} &= \frac{1}{2}m_A^2(1 - \cos 2\beta) + \frac{1}{2}m_Z^2(1 + \cos 2\beta), \\ (M_{\text{Re}\varphi}^2)_{12} &= (M_{\text{Re}\varphi}^2)_{21} = -\frac{1}{2}(m_A^2 + m_Z^2)\sin 2\beta, \\ (M_{\text{Re}\varphi}^2)_{22} &= \frac{1}{2}m_A^2(1 + \cos 2\beta) + \frac{1}{2}m_Z^2(1 - \cos 2\beta). \end{aligned} \quad (28.5.27)$$

Solving the secular equation gives the eigenvalues

$$m_H^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 + \sqrt{(m_A^2 + m_Z^2)^2 - 4m_A^2 m_Z^2 \cos^2 2\beta} \right], \quad (28.5.28)$$

$$m_h^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 - \sqrt{(m_A^2 + m_Z^2)^2 - 4m_A^2 m_Z^2 \cos^2 2\beta} \right]. \quad (28.5.29)$$

To calculate the masses of the charged scalars, we evaluate the potential V with the neutral scalars set equal to their vacuum expectation values:

$$\mathcal{H}_1 = \begin{pmatrix} v_1 \\ \mathcal{H}_1^- \end{pmatrix}, \quad \mathcal{H}_2 = \begin{pmatrix} \mathcal{H}_2^+ \\ v_2 \end{pmatrix}. \quad (28.5.30)$$

Using this in Eq. (28.5.10) then gives the quadratic part of the charged scalar potential

$$\begin{aligned} V_{\text{quad}}^C &= \frac{g^2}{2} |v_2(\mathcal{H}_1^-)^* + v_1\mathcal{H}_2^+|^2 + \frac{g^2 + g'^2}{4}(v_1^2 - v_2^2)(|\mathcal{H}_1^-|^2 - |\mathcal{H}_2^+|^2) \\ &\quad + (m_1^2 + |\mu|^2)|\mathcal{H}_1^-|^2 + (m_2^2 + |\mu|^2)|\mathcal{H}_2^+|^2 + B\mu\mathcal{H}_1^-\mathcal{H}_2^+. \end{aligned} \quad (28.5.31)$$

Using Eqs. (28.5.22) and (28.5.24), this may be written

$$\begin{aligned} V_{\text{quad}}^C &= \frac{1}{2}(m_W^2 + m_A^2)[|\mathcal{H}_1^-|^2(1 - \cos 2\beta) + |\mathcal{H}_2^+|^2(1 + \cos 2\beta) \\ &\quad + 2\sin 2\beta \mathcal{H}_1^-\mathcal{H}_2^+], \end{aligned} \quad (28.5.32)$$

where m_W is the charged gauge boson mass:

$$m_W^2 = \frac{1}{2}g^2(|v_1|^2 + |v_2|^2). \quad (28.5.33)$$

The charged scalar mass matrix is then

$$M_C^2 = \frac{1}{2}(m_W^2 + m_A^2) \begin{pmatrix} 1 - \cos 2\beta & \sin 2\beta \\ \sin 2\beta & 1 + \cos 2\beta \end{pmatrix}. \quad (28.5.34)$$

This has determinant zero, so it has one eigenvalue equal to zero and the other eigenvalue equal to the trace

$$m_C^2 = m_W^2 + m_A^2. \quad (28.5.35)$$

The zero-mass charged scalar is of course the other Goldstone boson associated with the spontaneous breakdown of $SU(2) \cdot U(1)$, and like the neutral Goldstone boson found earlier it is eliminated by the Higgs mechanism.

Even without knowing the parameters m_A and β , these results tell us a lot about the relative magnitudes of the scalar boson masses. We can rewrite Eqs. (28.5.28) and (28.5.29) in the form

$$m_H^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 + \sqrt{(m_A^2 - m_Z^2)^2 + 4m_A^2 m_Z^2 \sin^2 2\beta} \right], \quad (28.5.36)$$

$$m_h^2 = \frac{1}{2} \left[m_A^2 + m_Z^2 - \sqrt{(m_A^2 - m_Z^2)^2 + 4m_A^2 m_Z^2 \sin^2 2\beta} \right]. \quad (28.5.37)$$

We see that *the heavier neutral scalar mass m_H is larger than the larger of m_Z and m_A , while the lighter neutral scalar mass m_h is smaller than the smaller of m_Z and m_A* . If the large ratio of top to bottom quark masses is due to a large ratio $v_2/v_1 = \tan \beta$ of scalar field vacuum expectation values, rather than a large ratio of Yukawa couplings, then we expect β to be near $\pi/2$, in which case these inequalities become approximate equalities. Further, Eq. (28.5.35) shows that *the charged scalar mass is greater than both m_A and m_W* .

These results are quantitatively modified by various radiative corrections *within* the standard model (as opposed to radiative effects that produce the input parameters m_i^2 in theories of gauge-mediated supersymmetry breaking). The most important corrections arise from the presence in the scalar potential V of terms arising from graphs consisting of a single loop of top or bottom quarks interacting any number of times with external scalar field lines. This is because the top and bottom quarks have by far the strongest couplings to $\mathcal{H}_2$ and $\mathcal{H}_1$, respectively. (It is prudent to include bottom as well as top quark loops here because, as mentioned above, the larger mass of the top quark may be due to a large ratio v_2/v_1 , rather than to a large ratio of Yukawa couplings, but even in this case we will see that the dominant corrections are due to top quark loops.)

Let us first consider the neutral scalars, at least one of which would be lighter than the Z boson in the absence of radiative corrections. The effect of these top and bottom loops is to contribute a term of form $U_t(|\mathcal{H}_2^0|^2) + U_b(|\mathcal{H}_1^0|^2)$ to V^N . We will absorb any terms in

U_b or U_t that are linear in $|\mathcal{H}_1^0|^2 - v_1^2$ or $|\mathcal{H}_2^0|^2 - v_2^2$ into the input parameters m_1^2 and m_2^2 , so that

$$U'_b(v_1^2) = U'_t(v_2^2) = 0. \quad (28.5.38)$$

Our earlier results (28.5.24) and (28.5.22) for $m_1^2 + |\mu|^2$, $m_2^2 + |\mu|^2$ and $B\mu$ are then unchanged. Also, the mass matrix for the C -odd neutral scalars is still given by Eq. (28.5.26). On the other hand, the elements of the mass-squared matrix for the C -even neutral scalars are now given by

$$\begin{aligned} (M_{\text{Re}\varphi}^2)_{11} &= \frac{1}{2}m_A^2(1 - \cos 2\beta) + \frac{1}{2}m_Z^2(1 + \cos 2\beta) + \Delta_b, \\ (M_{\text{Re}\varphi}^2)_{12} &= (M_{\text{Re}\varphi}^2)_{21} = -\frac{1}{2}(m_A^2 + m_Z^2)\sin 2\beta, \\ (M_{\text{Re}\varphi}^2)_{22} &= \frac{1}{2}m_A^2(1 + \cos 2\beta) + \frac{1}{2}m_Z^2(1 - \cos 2\beta) + \Delta_t, \end{aligned} \quad (28.5.39)$$

where

$$\Delta_b = 2v_1^2 U''_b(v_1^2), \quad \Delta_t = 2v_2^2 U''_t(v_2^2). \quad (28.5.40)$$

The solutions of the secular equation are then

$$\begin{aligned} m_H^2 &= \frac{1}{2} \left[m_A^2 + m_Z^2 + \Delta_t + \Delta_b \right. \\ &\quad \left. + \sqrt{((m_A^2 - m_Z^2)\cos 2\beta + \Delta_t - \Delta_b)^2 + (m_A^2 + m_Z^2)^2 \sin^2 2\beta} \right], \end{aligned} \quad (28.5.41)$$

$$\begin{aligned} m_h^2 &= \frac{1}{2} \left[m_A^2 + m_Z^2 + \Delta_t + \Delta_b \right. \\ &\quad \left. - \sqrt{((m_A^2 - m_Z^2)\cos 2\beta + \Delta_t - \Delta_b)^2 + (m_A^2 + m_Z^2)^2 \sin^2 2\beta} \right]. \end{aligned} \quad (28.5.42)$$

In considering searches for these particles, it is important to note that the light Higgs mass m_h increases as the unknown mass m_A increases, reaching a finite upper bound for $m_A \rightarrow \infty$

$$m_h \leq m_h(m_A \rightarrow \infty) = m_Z^2 \cos^2 2\beta + \Delta_t \sin^2 \beta + \Delta_b \cos^2 \beta. \quad (28.5.43)$$

To calculate Δ_b and Δ_t , we recall from Section 16.2 that the potentials U_b and U_t are given by

$$U_b(|\mathcal{H}_1^0|^2) = -\frac{3}{16\pi^2} |\lambda_b \mathcal{H}_1^0|^4 \left[\ln \frac{|\lambda_b \mathcal{H}_1^0|^2}{M_{sb}^2} - \frac{3}{2} \right] + \text{linear terms}, \quad (28.5.44)$$

$$U_t(|\mathcal{H}_2^0|^2) = -\frac{3}{16\pi^2} |\lambda_t \mathcal{H}_2^0|^4 \left[\ln \frac{|\lambda_t \mathcal{H}_2^0|^2}{M_{st}^2} - \frac{3}{2} \right] + \text{linear terms}, \quad (28.5.45)$$

where $\lambda_t = m_t/v_2$ and $\lambda_b = m_b/v_1$ are the Yukawa couplings of the top and bottom quarks; M_{st} and M_{sb} are the masses of the stop and sbottom (the scalar superpartners of the top and bottom quarks); these masses and the terms $-3/2$ inside the square brackets are chosen to satisfy the condition that supersymmetry-breaking corrections due to stop and sbottom

loops would cancel the corrections due to top and bottom loops if the masses were equal; and the ‘linear terms’ are linear in $|\mathcal{H}_2^0|^2$ or $|\mathcal{H}_1^0|^2$, with coefficients adjusted to satisfy Eq. (28.5.38). (The factor 3 takes account of the three quark colors.) Then Eq. (28.5.40) gives

$$\Delta_b = -\frac{3}{4\pi^2}|\lambda_b|^4 v_1^2 \ln\left(\frac{\lambda_b^2 v_1^2}{M_{sb}^2}\right) = \frac{3\sqrt{2}m_b^4 G_F}{2\pi^2 \cos^2 \beta} \ln\left(\frac{M_{sb}^2}{m_b^2}\right), \quad (28.5.46)$$

$$\Delta_t = -\frac{3}{4\pi^2}|\lambda_t|^4 v_2^2 \ln\left(\frac{\lambda_t^2 v_2^2}{M_{st}^2}\right) = \frac{3\sqrt{2}m_t^4 G_F}{2\pi^2 \sin^2 \beta} \ln\left(\frac{M_{st}^2}{m_t^2}\right), \quad (28.5.47)$$

where $G_F = 1.17 \times 10^{-5} \text{ GeV}^{-2}$ is the Fermi coupling constant, given by Eq. (21.3.34) as $G_F = g^2/(4\sqrt{2}m_W^2)$. Taking $m_b = 4.3 \text{ GeV}$, $m_t = 180 \text{ GeV}$, $M_{st} \sim M_{sb} \sim 1 \text{ TeV}$, and $m_Z = 91.2 \text{ GeV}$ gives $\Delta_b \sim 1.1 \times 10^{-6} m_Z^2 / \cos^2 \beta$ and $\Delta_t \sim 1.1 m_Z^2 / \sin^2 \beta$. We see that even if $\tan \beta$ is as large as m_t/m_b , the top-quark correction Δ_t will still be much larger than Δ_b .

The effect of Δ_t is to *increase* both m_H and m_h . Taking this and other radiative corrections into account [26], for $\tan \beta > 10$ the upper bound (28.5.43) on the lightest neutral scalar mass is raised by radiative corrections from just below m_Z to between 100 GeV and 110 GeV for stop masses between 300 GeV and 1 TeV. For comparison, an experimental *lower* bound [27] of 62.5 GeV on m_h , m_H , and m_A is set by the absence of an hA or HA final state in e^+e^- collisions at 130 to 172 GeV. Also, calculations of radiative corrections involving Higgs scalars are consistent with precision measurements of electroweak phenomena for m_h in the range [27a] of 27 to 140 GeV.

Radiative corrections are less important for the charged scalars. Since the attachment of charged scalar lines allows transitions between top and bottom quarks, the correction to the scalar field potential here takes a more general form, constrained by $SU(2) \cdot U(1)$ to be

$$\Delta V = U\left(\mathcal{H}_2^\dagger \mathcal{H}_2, \mathcal{H}_1^\dagger \mathcal{H}_1, \mathcal{H}_2^\dagger \mathcal{H}_1, \mathcal{H}_1^\dagger \mathcal{H}_2, \mathcal{H}_1^T e \mathcal{H}_2\right). \quad (28.5.48)$$

(Quark loops do not actually produce any dependence on $\mathcal{H}_1^T e \mathcal{H}_2$.) Every appearance of a $\mathcal{H}_1$ or $\mathcal{H}_2$ doublet is accompanied with a factor λ_b or λ_t , respectively, so terms involving $\mathcal{H}_1$ will be suppressed, as we have already seen in calculating the neutral scalar masses. To a good approximation, then, the correction to the effective potential is of the form

$$\Delta V \simeq U(\mathcal{H}_2^\dagger \mathcal{H}_2, 0, 0, 0) = U(|v_2 + \varphi_2|^2 + |\mathcal{H}_2^-|^2, 0, 0, 0). \quad (28.5.49)$$

By going back to the case where the charged fields vanish, we see that the function U must be just the same as what was earlier called U_t . Any terms in U of first order in an expansion in powers of $|v_2 + \varphi_2|^2 + |\mathcal{H}_2^-|^2 - v_2^2$ would simply serve to redefine the constant m_2^2 , and are eliminated by the convention (28.5.38). Terms in U of second order in $|v_2 + \varphi_2|^2 + |\mathcal{H}_2^-|^2 - v_2^2$ are genuine radiative corrections, but although they contain terms of second order in $|\varphi_2|^2$ which do affect the neutral scalar masses, they contain no terms of second order in $|\mathcal{H}_2^-|^2$ which could shift the non-Goldstone charged scalar mass. Fortunately, radiative corrections are not needed to avoid a conflict with experiment, because in the absence of an upper bound on m_A there is no theoretical upper bound on the charged scalar mass (28.5.35).

An experimental lower bound [28] $m_C \geq 59$ GeV is provided by the non-observation of the process $e^+e^- \rightarrow \mathcal{H}^+\mathcal{H}^-$ at 181 to 184 GeV. There is a much more stringent lower bound on m_C provided by the rate of the process $b \rightarrow s\gamma$ (measured in decays like $B \rightarrow K^*\gamma$), which can occur by transitions to an intermediate $\mathcal{H}^-u$ or $\mathcal{H}^-c$ state, with the photon radiated from the virtual quark or $\mathcal{H}^-$. The present agreement between theory and experiment for this process sets a lower bound [29] of about 150 GeV on m_C (and higher for $\tan\beta < 1$). Using Eq. (28.5.35), this gives an important lower bound $m_A > 125$ GeV.

There are two conditions on the masses m_i^2 that must be satisfied by any model of supersymmetry breaking in order to give a successful account of electroweak symmetry breaking. One of them is provided by the requirement that the potential is bounded below, which as we have seen requires that

$$2|\mu|^2 + m_1^2 + m_2^2 > B\mu.$$

Since $B\mu$ is defined to be positive, this ensures that the C -odd neutral scalar squared mass (28.5.21) is positive. The other condition is provided by Eqs. (28.5.22) and (28.5.24), which for arbitrary values of β require that

$$4(m_1^2 + |\mu|^2)(m_2^2 + |\mu|^2) \leq (B\mu)^2. \quad (28.5.50)$$

We can easily see from Eq. (28.5.10) that this condition ensures that the matrix of second derivatives of the potential has a negative eigenvalue at $\mathcal{H}_1 = \mathcal{H}_2 = 0$, so this $SU(2) \cdot U(1)$ -invariant point is one of unstable equilibrium, and therefore $SU(2) \cdot U(1)$ must be spontaneously broken. If β is very near $\pi/2$, then Eq. (28.5.24) tells us that this condition is satisfied by having $m_1^2 + |\mu|^2$ positive and $m_2^2 + |\mu|^2$ *negative*. As we will see in the next section, the renormalization group flow of the parameters in the scalar field Lagrangian provides a mechanism for driving $m_2^2 + |\mu|^2$ negative.

Even with a minimal set of superfields, in supersymmetry theories there are several pairs of particles with different $SU(2) \cdot U(1)$ transformation properties, but the same charge, color, and baryon and lepton number, which become mixed when $SU(2) \cdot U(1)$ is spontaneously broken. We have already seen an example of this in the previous section, in which we had to deal with the mixing of the scalar superpartners of the left-handed quarks with the complex conjugates of the scalar superpartners of the left-handed antiquarks. A similar mixing occurs between the higgsinos and gauginos, both charged and neutral; the particles of definite mass are not higgsinos or gauginos, but mixtures known as *charginos* and *neutralinos*. Let us consider the charginos, which provide a useful bound on μ . According to Eq. (27.4.8), there are off-diagonal supersymmetric mass terms in the Lagrangian density

$$-\text{Re} \left[\mu \left(h_{1L}^{-T} \epsilon h_{2L}^+ \right) + i\sqrt{2}m_W \cos\beta \left(w_L^{-T} \epsilon h_{2L}^+ \right) + i\sqrt{2}m_W \sin\beta \left(w_L^{+T} \epsilon h_{1L}^- \right) \right],$$

To this, we should add a term for the wino mass generated by gauge interactions with the supersymmetry-breaking sector

$$-m_{\text{wino}} \text{Re} \left(w_L^{+T} \epsilon w_L^- \right).$$

The squared chargino masses are then the eigenvalues of the matrix $\mathcal{M}_C^\dagger \mathcal{M}_C$, where

$$\mathcal{M}_C = \begin{pmatrix} m_{\text{wino}} & i\sqrt{2}m_W \sin \beta \\ i\sqrt{2}m_W \cos \beta & \mu \end{pmatrix}. \quad (28.5.51)$$

These two eigenvalues are

$$\begin{aligned} m_{\text{chargino}}^2 = \frac{1}{2} & \left[m_{\text{wino}}^2 + 2m_W^2 + |\mu|^2 \right. \\ & \pm \left((m_{\text{wino}}^2 - |\mu|^2)^2 + 4m_W^4 \cos^2 2\beta \right. \\ & \left. \left. + 4m_W^2 (m_{\text{wino}}^2 + |\mu|^2 - 2m_{\text{wino}} \operatorname{Re} \mu \sin 2\beta) \right)^{1/2} \right]. \end{aligned} \quad (28.5.52)$$

We expect the wino mass m_{wino} to be much larger than m_W . If it is also much larger than $|\mu|$, then the heavier chargino is mostly a wino, with mass m_{wino} , while the lightest chargino is mostly a higgsino, with mass $|\mu|$. In any case, $|\mu|$ is greater than the lightest chargino mass, which the non-appearance of gauginos in e^+e^- annihilation tells us is greater than about 60 GeV, and probably greater than m_W . The search for neutralinos in e^+e^- annihilation (with assumptions like (28.4.2)) has set a lower bound of 27 GeV on the mass of the lightest neutralino [29a].